

Urgent needs, opportunities and challenges of virtual reality in healthcare and medicine in the era of large language models

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Abstract: The convergence of large language models (LLMs) and virtual reality (VR) technologies has led to significant breakthroughs across multiple domains, particularly in healthcare and medicine. Owing to its immersive and interactive capabilities, VR technology has demonstrated exceptional utility in surgical simulation, rehabilitation, physical therapy, mental health, and psychological treatment. By creating highly realistic and

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precisely controlled environments, VR not only enhances the efficiency of medical training but also enables personalized therapeutic approaches for patients. The convergence of LLMs and VR extends the potential of both technologies. LLM-empowered VR can transform medical education through interactive learning platforms and address complex healthcare challenges using comprehensive solutions. This convergence enhances the quality of training, decision-making, and patient engagement, paving the way for innovative healthcare delivery. This study aims to comprehensively review the current applications, research advancements, and challenges associated with these two technologies in healthcare and medicine. The rapid evolution of these technologies is driving the healthcare industry toward greater intelligence and precision, establishing them as critical forces in the transformation of modern medicine.

Keywords: Large language model; Virtual reality; Healthcare; Medicine

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1 Introduction

Since its inception in the 1960s, virtual reality (VR) technology has advanced significantly, driven by rapid developments in computer graphics and artificial intelligence (AI). VR is a user–computer interface created through hardware and software that allows users to engage with multiple sensory systems, simulating environments that resemble the real world for interactive experiences [1]. By the late 1990s, with the widespread adoption of VR technology, VR-based rehabilitation therapies were increasingly integrated into healthcare systems [2]. VR has diverse applications in medical training, patient therapy, and surgical assistance. Several studies have demonstrated the efficacy of VR in addressing various clinical conditions, including acute pain [3], chronic obstructive pulmonary disease [4], and social anxiety disorders [5]. Furthermore, VR leverages a brain mechanism known as embodied simulation, enabling patients to interact with their environment through digital media in a manner that fosters both physical and emotional engagement, thereby enhancing their participation [6]. Moreover, the capability of VR to automatically track and record user performance offers exceptional flexibility and control in treatment tasks [7,8], allowing patients to undergo high-intensity training while maintaining an enjoyable and engaging therapeutic experience [9].

Language models (LM) are designed to model the probability of sequence generation with the primary task of estimating the probability distribution of the next word or character in a specific context. These models assess the linguistic plausibility of sentences and are widely used in various natural language processing (NLP) tasks. The origins of language modeling date back to the late 1980s with statistical language models [10–14], which used n-gram statistical methods to predict the likelihood of the next word in a given text, thereby enhancing the performance of tasks such as information retrieval [15]. Over the past decade, driven by advances in hardware such as graphics processing units and the exponential growth of data, pre-trained language models (PLMs) have emerged as the dominant research direction. As training datasets have continued to expand, model depth has increased, and parameter scales have grown significantly. Large language models (LLMs), defined by parameter counts exceeding billions, typically leverage the transformer architecture for modeling [16,17]. Foundation models, which are large-scale AI models pretrained on diverse data (text, images, and structured medical records), represent a paradigm shift by enabling transfer learning across healthcare tasks. In medicine, they unify fragmented data modalities

(e.g., electronic health records, imaging, and genomics) to support tasks from diagnosis to drug discovery, thereby reducing the need for task-specific training and accelerating innovation [18]. These models have demonstrated exceptional performance in areas such as question answering, machine translation, and text generation. Trained on vast corpora, LLMs represent a significant extension of parameter size over earlier PLMs, and their performance across tasks markedly improves as the model scale increases [19].

Recent advancements in LLMs have been rapid and transformative, such as Anthropic's Claude 3, Google DeepMind's Gemini Ultra, and Meta's Llama 3 pushing performance boundaries. These models outperform their predecessors in terms of medical reasoning and multilingual support. The field is evolving at an accelerating pace, with model capabilities doubling every 6–12 months owing to innovations in architecture (e.g., Mixture-of-Experts) and training efficiency. LLMs are powerful tools that increasingly help healthcare professionals manage and process information efficiently. As a type of generative AI, LLMs can accept image and text inputs and produce text outputs, thereby demonstrating significant potential in various aspects of healthcare.

The continuous development of LLMs and VR technologies has led to breakthroughs in various fields. This convergence has spanned intelligent dialogue systems, virtual character interactions, and virtual assistants in augmented reality. LLM-empowered VR, which combines data-driven analytics with immersive medical diagnostics, is emerging as a cutting-edge frontier in healthcare. This convergence is becoming a key focus in contemporary healthcare research.

This study provides a comprehensive review and analysis of the current developments in VRs and LLMs, as well as their convergence and applications in healthcare and medicine. In addition, it explores potential future developments in and challenges of LLM-empowered VR.

2 Virtual reality in healthcare and medicine

VR technology has demonstrated broad potential in the healthcare field, encompassing various areas such as surgical simulation, rehabilitation, physical training, and psychotherapy. Table 1 summarizes its specific applications.

Table 1 Summary of virtual reality in the healthcare field

Application area	Benefits	Examples
Real and simulated surgery	Helps surgeons practice procedures to anticipate complications, thereby improving accuracy and reducing errors in real surgeries	Transanal total mesorectal excision [22] Laparoscopic hysterectomy [23] Orthopedic trauma surgery [24] Percutaneous tumor puncture [25]
Rehabilitation and physical training	Helps restore motor coordination, muscle strength, and cognitive function	Stoke [29] Alzheimer's disease [30] Spinal cord injury [32] Physical training for athletes [34] Daily light exercise for older adults [35]
Psychological therapy and mental health	Provides controlled environments for practicing coping skills, social interactions, and trauma recovery	Anxiety and fear [37,41] Depressive symptoms [38] Social fear [39] PTSD [40]

2.1 Real and simulated surgery

Surgical simulation is a widely used training method that offers a learning environment designed to minimize patient risk [20]. VR-based surgical simulation not only helps surgeons familiarize themselves with procedures but also allows for the simulation of complex cases and unexpected situations, offering

more comprehensive skills training. For example, preoperative VR simulations enable surgeons to review a patient's anatomical structure and anticipate potential issues, thereby improving surgical success rates. Studies have shown that immersive and realistic experiences contribute to the acquisition of additional skills by medical professionals [21]. VR has been used in transanal total mesorectal excision [22], laparoscopic hysterectomy [23], orthopedic trauma surgery [24], and percutaneous tumor puncture [25]. However, many surgical simulators rely solely on VR environments without incorporating haptic feedback [26], resulting in low immersion in the simulation environment, which in turn results in quantitative feedback that does not provide an accurate representation of a surgeon's capabilities.

2.2 Rehabilitation therapy and physical training

Recently, VR technology has been extensively applied to enhance motor skills and facilitate patient rehabilitation. Studies have shown that VR can aid in restoring motor function, strengthening muscles, and improving joint mobility. VR enhances patient engagement by leveraging immersive environments, thereby increasing the effectiveness of rehabilitation training [27]. For example, VR is particularly beneficial for patients with stroke because it supports motor function recovery. Virtual exercises help patients coordinate hand and lower limb movements, thereby promoting motor skill enhancement and neurological recovery [28]. Compared to traditional therapies, VR significantly improves motor coordination and fosters greater autonomy in movement in patients with stroke [29]. For patients with Alzheimer's disease, VR cognitive training games help maintain brain activity and mitigate memory decline [30]. Additionally, VR offers older adults a novel approach to health management, thus improving their overall physical well-being [31]. VR is also highly effective for patients with spinal cord or brain injuries [32]. VR enables personalized training programs by creating tailored simulated environments, resulting in more efficient and meaningful rehabilitation outcomes [33].

VR technology has demonstrated significant advantages in physical fitness enhancement. Immersive virtual environments enable personalized training programs, thereby allowing athletes to engage in repetitive practice in simulated settings to improve their motor performance and skills [34]. VR also provides a safe training environment for older adults, who often face declining physical capabilities. It facilitates low-impact exercise and reduces the risk of injury while enhancing engagement in and the overall effectiveness of physical training [35].

VR-based motor rehabilitation training is most suitable for patients with stroke with essentially intact cognitive function, specifically those who are capable of understanding simple tasks and without severe sensory impairments (i.e., adequate visual, auditory, and vestibular function). This is particularly beneficial for individuals with high acceptance of new technologies or low adherence to traditional rehabilitation because the gamified nature of VR interventions enhances engagement. Notably, educational attainment does not impose strict limitations. The core determinant is the patient's ability to interact effectively with virtual scenarios and maintain consistent training compliance. This targeted applicability aligns with the interactive and immersive characteristics of VR systems, which rely on meaningful sensory-motor integration to drive functional recovery in post-stroke populations [36].

2.3 Psychotherapy and mental health

VR technology has significant potential in the field of psychotherapy beyond its applications in surgical simulation, rehabilitation, and physical training. As an adjunctive tool, VR can assist patients in addressing mental health conditions such as anxiety, depression, and post-traumatic stress disorder (PTSD). VR creates immersive virtual environments that enable patients to confront and manage emotional challenges, gradually alleviating their anxiety and fear while improving their overall emotional well-being [37]. For example, in

the treatment of depression, VR provides a virtual space that offers psychological comfort and emotional support, thus aiding the alleviation of depressive symptoms [38]. For patients with social anxiety, VR can simulate interactive social scenarios, allowing them to practice social skills in a controlled environment and reduce social fears [39]. Similarly, VR plays a critical role in PTSD therapy by recreating trauma-related memories, thereby offering patients a safe environment to regulate their emotions and facilitating psychological recovery [40]. Moreover, VR has significant advantages in addressing psychological issues in children. For example, children often experience heightened anxiety during the perioperative period, which can affect surgical outcomes and lead to long-term psychological effects. Studies have shown that VR tours of operating rooms effectively reduce preoperative anxiety in pediatric patients [41]. This innovative approach enhances psychological resilience and emotional regulation and offers a transformative tool for mental health treatment.

3 Large language models in healthcare and medicine

The recent rapid development of LLMs has driven transformative changes in healthcare institutions, fundamentally reshaping the manner in which medical researchers and clinicians interact with medical data, patients, and extensive knowledge repositories [42]. These models, with their human-like capabilities to understand and generate text, demonstrate significant potential in the medical domain [43]. The emergence of groundbreaking LMs has profoundly affected the field of NLP. Among these advancements, the introduction of transformer architecture marks a pivotal milestone, as it has enabled the development of PLMs such as the Bidirectional Encoder Representations from Transformers (BERT) and Robustly Optimized BERT Pretraining Approach (RoBERTa) [44,45]

BERT, introduced by Devlin et al. in 2018 [44], leverages deep bidirectional pretraining on large-scale corpora, thereby outperforming earlier models across multiple tasks and driving significant advancements in NLP. RoBERTa, proposed by Liu et al. in 2019 [45], enhances BERT's performance through optimized pretraining techniques, thus establishing new benchmarks in the field.

LLMs are increasingly utilized in the healthcare industry, demonstrating significant value across domains such as clinical decision support, medical education and research, patient health management, and consultation. Their widespread adoption not only enhances diagnostic accuracy but also improves the efficiency of healthcare services. This section provides an in-depth exploration of LLM applications in healthcare.

3.1 Clinical decision support

Clinical data are inherently complex and diverse and require physicians to develop personalized diagnostic and treatment plans tailored to each patient's specific needs. Leveraging the robust predictive and classification capabilities of LLMs can facilitate the integration of individualized patient information with extensive medical knowledge to provide customized clinical decision support. LLMs have demonstrated remarkable potential for aiding diagnosis and early disease screening. For example, the Harvard Medical School team introduced PathChat, a general-purpose visual-language AI assistant trained on over one million biopsy image fragments using self-supervised learning, thereby enabling precise disease identification from pathology slides [46]. The introduction of LLMs has significantly improved diagnostic accuracy and predictive capabilities, facilitating early disease detection and the creation of personalized treatment plans. LLMs can analyze patient symptoms, medical histories, and clinical data to identify diseases early, thus enabling timely intervention [47,48]. For example, transformer-based models such as BERT, RoBERTa, and DistilBERT have been used to predict sensory changes in COVID-19 patients [49].

Researchers at Harvard University developed the Clinical Histopathology Imaging Evaluation Foundation (CHIEF), a multimodal foundation model integrating vision and language, capable of diagnosing 19 types of cancers. Unlike traditional LLMs that process text exclusively, multimodal models such as CHIEF and PathChat analyze images and text, enabling cross-modal reasoning, which is critical for histopathology [50]. Furthermore, studies have explored the application of LLMs in diagnosing Alzheimer's disease [51] and dementia [52]. LLMs have also demonstrated significant potential in diabetes management, offering substantial contributions to improving treatment strategies [17].

3.2 Medical education and research

Medical education and research encompass an extensive range of knowledge domains, and LLMs have proven invaluable in facilitating efficient access to and understanding of complex medical concepts. By leveraging NLP technologies, LLMs can rapidly generate educational materials, provide intelligent question-answering systems, and offer personalized learning recommendations tailored to student needs. This can facilitate a more individualized, precise, and efficient approach to medical education. Additionally, LLMs can support educators by assisting with teaching evaluations and ensuring that students accurately comprehend key concepts.

In medical research, LLMs excel at processing diverse data sources, including medical literature, clinical records, and pathology reports. They effectively perform tasks such as text summarization, entity recognition, and relationship extraction, thereby enabling researchers to extract meaningful insights from large datasets. This automation significantly accelerates the accumulation and dissemination of medical knowledge, enhancing collaborative and evidence-based research efforts [53,54].

LLMs also have a profound impact on drug development. They can identify therapeutic compounds by analyzing complex molecular structures and their efficacy and safety with remarkable efficiency [55]. For example, the TxGNN model, trained on a medical knowledge graph encompassing 17,080 disease mechanisms and 7,957 drug action mechanisms, has successfully identified potential treatments for diseases with limited therapeutic options and sparse molecular data [56]. Furthermore, DrugFormer was developed to integrate sequenced gene markers and gene-based knowledge graphs to address drug resistance, which is an ongoing challenge in chemotherapy and targeted therapies. This model can achieve high precision in predicting drug resistance at the single-cell level, as demonstrated by comprehensive analyses of single-cell data from various cancer types. DrugFormer has also proven effective in identifying cell types and uncovering the underlying molecular mechanisms [57].

3.3 Patient management and health consultation

In addition to diagnostics and medical research, LLMs play a pivotal role in patient care. They empower healthcare providers to design personalized care plans, thus enhancing both the quality of care and patient experience [58,59]. Moreover, LLM-driven chatbots have been widely adopted for patient consultation and health management. For example, the SSPEC triage model, trained on a knowledge base derived from 35,418 real-world triage dialogues, has demonstrated superior factual accuracy, safety, and empathetic capabilities. Compared with human triage services, the SSPEC effectively reduces medical disputes while significantly improving triage efficiency [60]. In summary, LLMs have demonstrated immense potential for optimizing patient management and healthcare processes, driving greater efficiency in medical services, and increasing the quality of patient care.

4 Development and applications of LLM-empowered VR

The continuous development of LLMs and VR technologies has recently led to breakthroughs in many

fields. The convergence of these two technologies offers users a more intelligent, personalized, and immersive interactive experience.

4.1 Interactive intelligent dialogue systems

LLM-empowered VR enables the creation of more intelligent dialogue systems in virtual environments. LLM can empower virtual characters with NLP capabilities, making the interactions between users and virtual characters more natural and fluid. For example, users can engage in conversations with a virtual guide during a virtual tour to receive real-time travel information and recommendations, enhancing immersion [61]. This can enrich and diversify users' interactive VR experiences.

The technical integration of LLMs and VR enables real-time NLP via edge-deployed lightweight models (e.g., DistilBERT), achieving < 500 ms response times critical for immersion. In surgical VR, multimodal embeddings (e.g., CAT-ViL in Surgical-VQLA) fuse video frames with procedural text, allowing virtual mentors to answer queries such as, "Where is the hepatic artery?" by localizing structures in real time [62].

4.2 Intelligent training and assistance systems

LLM-empowered VR contributes to intelligent training and can be applied to assistance systems. VR provides learners with an immersive training environment in fields such as medical training and education. Meanwhile, LLM can answer learners' questions and provide guidance in real time [63].

4.3 Enhanced user experience and personalized recommendations

By combining LLMs and VR, systems can offer personalized virtual experiences based on user requirements and preferences. For example, in virtual shopping, users can converse with virtual sales representatives to obtain product recommendations and descriptions. In virtual education, the system can tailor learning content according to students' progress and interests. This personalized recommendation enhances user engagement and satisfaction, thereby expanding VR application scenarios.

5 Applications and future outlook of LLM-empowered VR in healthcare and medicine

LLM-empowered VR has revolutionized the healthcare sector by offering users unprecedented immersive experiences. As this technology evolves, it has significant potential for enhancing healthcare quality and improving patient outcomes, ultimately playing a pivotal role in the transformation of future medical practices.

One study introduced a transformer model with CAT-ViL embeddings for surgical Visual Question Localization Answering (VQLA) tasks. This model provides localized answers tailored to specific surgical scenarios and related queries, representing an important step in the development of VQLA systems for surgical training and scene understanding. The VQLA system facilitates the learning and comprehension of surgical procedures among medical students and junior surgeons from recorded surgical videos [63]. Researchers have also proposed an end-to-end trainable language-vision GPT model (LV-GPT) designed to address visual question answering (VQA) tasks during surgery. By extending the GPT-2 model and incorporating visual inputs, LV-GPT integrates a visual feature extractor and visual token embeddings to merge image and language information effectively. The model employs a carefully crafted token sequence that arranges language and visual tokens to simulate human comprehension of the task. Experimental results have demonstrated that LV-GPT outperforms existing state-of-the-art models across multiple publicly available surgical VQA datasets [64].

In the future, LLM-empowered VR will showcase extensive applications in healthcare, becoming an

essential tool and resource. VR has significant potential for use in healthcare, particularly in physical rehabilitation, therapy, and psychological treatment. VR increases patient engagement and improves rehabilitation outcomes by enhancing immersion, interactivity, and customization. Furthermore, the incorporation of LLMs will only amplify the advantages of VR technology. As Figure 1 shows, the integration of LLMs and VR has numerous potential medical applications. First, this fusion will revolutionize medical education and training. By combining LLMs and VR, medical education can provide more personalized and immersive learning experiences. LLMs can provide real-time answers, knowledge reinforcement, and diagnostic guidance, while VR can create simulated surgical and clinical environments, thereby enabling students to practice in a risk-free setting. This integration will substantially improve the efficiency and quality of medical education. Second, LLM-empowered VR will enhance personalized clinical decision support. LLMs can process complex medical data to provide personalized diagnostic and treatment recommendations. When paired with VR, doctors can visually explore a patient's anatomy and lesion locations in a virtual environment, thereby enabling more accurate clinical decisions. This fusion helps increase treatment success rates and reduce medical errors. Furthermore, LLM-empowered VR can transform surgical simulation and training. LLMs assist doctors in understanding the complexities of surgical decision-making, while VR allows them to rehearse surgeries in a controlled virtual environment. This not only improves technical skills but also enhances the ability of doctors to handle unexpected situations during real procedures. Moreover, the integration of VR and LLMs can enhance the patient experience and psychological health treatment. VR applications extend beyond physical rehabilitation and offer a powerful tool for addressing psychological health issues. LLMs can provide emotional support and real-time feedback, thereby helping patients cope with anxiety, depression, and PTSD in virtual settings and delivering personalized treatment plans. Finally, LLM-empowered VR makes remote healthcare and health management more feasible. Doctors can interact with patients in real time via VR for remote diagnosis and treatment, and LLMs can analyze symptoms and provide personalized health recommendations to assist patients in daily health management.

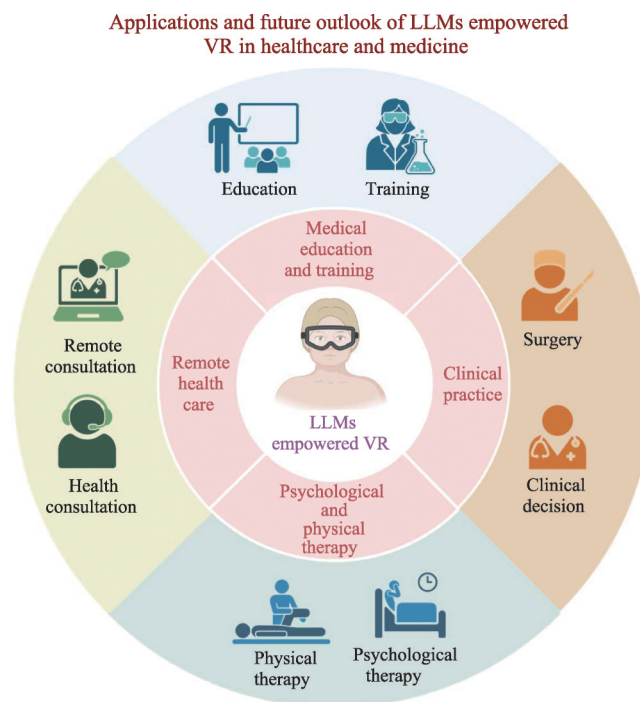


Figure 1 Applications and future outlook of LLMs empowered VR in healthcare and medicine.

The efficacy of these technologies has been evaluated using multidimensional metrics. For example, in surgical training, criteria include laparoscopic suturing task completion time (<10 minutes), error margin (< 2 mm deviation from targets), and skill transfer accuracy to physical phantoms (>85% concordance with expert performance) [65]. Rehabilitation outcomes are measured by improvements in Fugl-Meyer Assessment scores (≥ 5 points post-intervention), patient adherence (>80% session attendance), and caregiver-reported functional gains, such as independent feeding [66]. LLM-based diagnostics are assessed through concordance with specialist opinions (>90%) and reduced consultation duration (mean 15 ± 3 minutes versus 30 ± 5 minutes for traditional reviews) [67].

Continued research and application will facilitate further exploration of the potential of these technologies, leading to breakthroughs and innovations that will bring substantial benefits to the medical field, improve healthcare delivery and efficiency, and ultimately benefit global patients and the medical community.

6 Challenges of LLM-empowered VR in healthcare and medicine

Although LLM-empowered VR holds immense potential in the medical field, it also presents significant challenges that require further research and resolutions. Figure 2 outlines the key challenges and prospects of this integration.

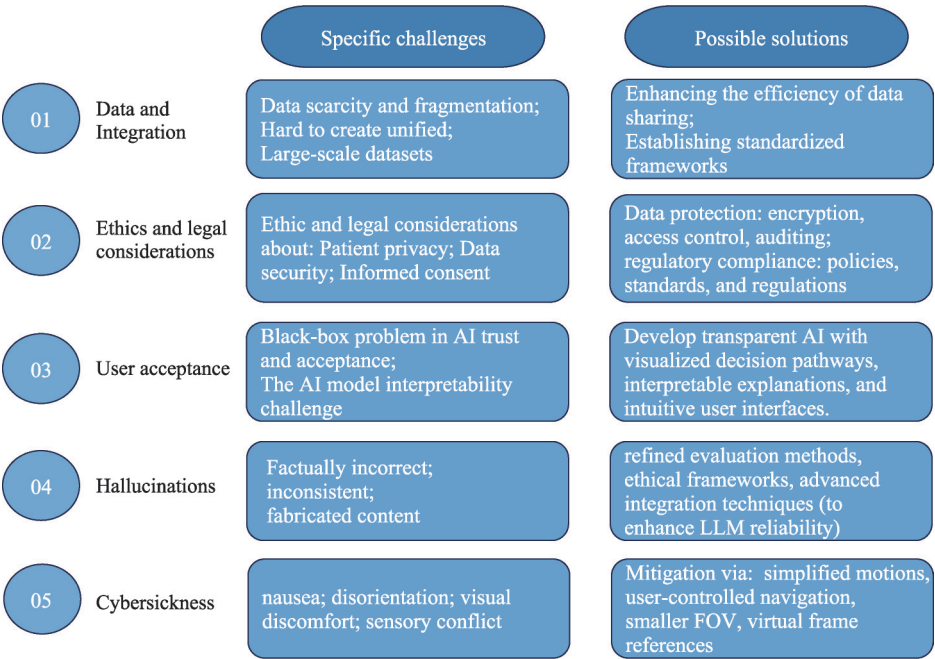


Figure 2 Challenges of LLMs Empowered VR in Healthcare and Medicine. They include data and integration, ethics and legal considerations, user acceptance, hallucinations and cybersickness.

6.1 Data and integration

High-quality diverse datasets are fundamental for training high-performance AI models. However, AI-driven medical VR (AI-VR) technologies face challenges related to data scarcity and fragmentation. Medical data acquisition is constrained by geographic, institutional, and regulatory limitations, making it difficult to create unified large-scale datasets. Furthermore, variations in data formats, standards, and quality across sources hinder effective integration and limit the consistency of support provided to AI models. Moreover, integrating AI-VR technologies into existing healthcare systems is a complex and challenging task that

requires addressing technical compatibility issues while ensuring that the stability and security of the current systems remain unaffected during the integration process. To overcome these challenges, future efforts should focus on establishing standardized frameworks for medical data integration, including (1) transfer learning (TL) to reduce target data reliance; (2) active learning (AL) to select key samples for expert annotation and boost performance cheaply; (3) one-shot/few-shot learning (OSL) to extract patterns from minimal samples for rare targets; (4) multi-task learning (MTL) to train related tasks jointly, aiding with limited data; (5) data augmentation (DA) to transform data to increase diversity and reduce overfitting; (6) data synthesis (DS) to generate realistic data via models to supplement scarcity; and (7) federated learning (FL) to enable secure cross-institutional training [68].

6.2 Ethics and legal considerations

Ethical and legal challenges are central to the application of AI in healthcare, particularly those regarding patient privacy, data security, and informed consent. AI-VR systems must ensure robust patient privacy protections and safeguard the collection, storage, and usage of medical data. Measures such as data encryption, access control, and auditing mechanisms are essential for mitigating the risk of sensitive information breaches. Furthermore, the practical implementation of AI-VR technology may involve errors or unforeseen incidents, raising questions about accountability. The lack of clear legal guidelines regarding responsibility creates significant challenges for healthcare providers, technology developers, and patients. Therefore, to reduce legal disputes and risks, comprehensive regulations and industry standards that define the roles and responsibilities of all stakeholders in AI-VR applications must be established.

6.3 User acceptance

The “black box” nature of AI systems is a significant barrier to user trust and acceptance. Healthcare professionals and patients often seek to understand the decision-making processes involved in AI-assisted diagnostics. However, the complexity of current AI models makes their reasoning challenging to explain, thereby potentially undermining user confidence. Researchers must address this issue by developing more transparent AI systems capable of visualizing decision pathways and generating easily interpretable explanations.

In addition, designing user-friendly and intuitive interfaces is critical. Simplified designs can alleviate the fears associated with new technologies and enhance their usability for clinicians and patients. For diverse user groups, providing personalized interaction experiences is essential for promoting widespread adoption. These efforts will enable the seamless integration of AI-VR technologies into clinical practice, thereby fostering greater trust and engagement.

6.4 Hallucinations in clinical LLMs

Hallucinations in LLMs refer to the generation of seemingly coherent content that is factually incorrect, inconsistent, or entirely fabricated, and represent a critical challenge for their application. It encompasses four primary types: intrinsic hallucinations stemming from internal knowledge limitations, extrinsic hallucinations arising from the flawed integration of external information, amalgamated hallucinations resulting from the incorrect fusion of multiple facts, and non-factual hallucinations that contradict established knowledge.

The underlying mechanisms include knowledge overshadowing, insufficient knowledge representation, failures in information extraction, contextual misalignment, and semantic entropy, along with contributing factors, such as training data quality, model architecture, task complexity, and dynamic reasoning processes.

Hallucinations pose severe risks in critical domains, such as healthcare and scientific research, including

misdiagnoses and erroneous conclusions. Conversely, they may foster creativity in fields such as art by generating unconventional ideas. Detection methods involve named entity recognition and probability-based approaches, while mitigation strategies include prompt engineering, fine-tuning, and retrieval-augmented generation. Future efforts should prioritize refined evaluation methods, ethical frameworks, and advanced integration techniques to enhance the reliability of LLMs.

6.5 Cybersickness in virtual reality

Cybersickness in VR is a motion-sickness-like syndrome triggered by visual stimuli and presents with symptoms such as nausea, disorientation, and visual discomfort. Its core mechanism stems from sensory conflict, reflecting the mismatch between VR visual inputs and signals from the vestibular and somatosensory systems, which disrupts the brain's ability to coordinate spatial perception. Key influencing factors include VR content attributes. Complex camera movements (e.g., roll rotations, rapid translations, and acceleration) exacerbate symptoms, which can be mitigated by simplified motions or user-controlled navigation. Other critical factors are field of view (a smaller field of view reduces severity), virtual frame references (e.g., “virtual nose”), and exposure duration (longer sessions increase risk).

Regarding individual variability, users over 30 exhibit higher susceptibility in controllable scenes, potentially because of their reduced interaction proficiency. Motion sickness susceptibility (assessed using MSSQ scores) is significantly correlated with symptom severity, although no gender-based differences have been observed.

7 Conclusion

The technological applications of LLMs and VR are reshaping healthcare delivery paradigms. With its immersive interactivity, VR creates quantifiable and highly controllable medical environments for surgical simulation, neurorehabilitation, and psychological interventions. Meanwhile, LLMs provide intelligent support for clinical decision-making and medical education through in-depth analyses of healthcare data and natural language interaction capabilities. Their convergence not only achieves technical synergy, such as LLMs generating real-time personalized guidance in VR simulations and dynamically adjusting treatment plans during rehabilitation, but also drives the transition from standardized to precision healthcare, creating more efficient and human-centered interactions between clinicians and patients.

However, this integrative pathway continues to face substantial challenges. At the data level, fragmented medical datasets and privacy protection requirements constrain the training efficacy of LLM-empowered VR systems. Technically, VR-induced cybersickness and LLM hallucination risks lack universal solutions. Ethically, liability frameworks for AI-assisted decisions and regulatory guidelines for virtual healthcare scenarios remain underdeveloped. Addressing these issues requires interdisciplinary collaboration between computer science, clinical medicine, and ethics to establish a model that balances innovation with risk mitigation.

Future advancements in LLM-empowered VR healthcare applications will evolve toward “deep synergy” and “scenario democratization”. Multimodal foundation models will enhance perceptual integration, such as LLM-empowered surgical VR systems that analyze intraoperative imaging and clinician commands to generate dynamic risk alerts. Furthermore, standardized data-sharing protocols and lightweight devices will facilitate penetration into primary care and remote rehabilitation. Ultimately, technological progress must remain anchored in healthcare's core mission of patient-centricity. Enhancing efficiency while upholding medical safety and humanistic care constitutes the pivotal value of LLM-empowered VR in reshaping healthcare ecosystems.

Declaration of competing interest

All authors declare there are no competing interests.

CrediT authorship contribution statement

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